Brief for the Head of Fire Services: Hazira gas terminal vapour release and fire, 24 September 2020 (draft)

Learning purposes: not a statutory document. Derived from a ProbeMatrix investigation export; learnings are evidence-linked but should be validated by the organisation before adoption.

What happened

In the early hours of **24 September 2020**, condensate began leaking at **02:12:50** from a 1986 meter on the Gas Terminal metering skid at the Hazira Gas Processing Complex. The meter's gasket had been replaced six days earlier with a locally bought O-ring, and the repair had not been tested to design pressure. A dense vapour cloud formed. **CISF guards saw and heard it at 02:15** and informed the **Fire Station**, among others. The control room that owned the equipment learned of it 3 to 5 minutes later, through the CCTV room and a gas-detection alarm that only displayed on a local panel.

Engineers and then the shift in-charge team tried to reach the leak and were turned back by the cloud. At about 02:35 the Fire Station was told to stop all vehicle movement in the plant. The emergency plan named the major-leak scenario but gave **no isolation criteria and no delegated authority**, and the isolation valves could only be closed by hand. The first valve was closed at **03:01, 46 minutes after detection**. The cloud **ignited at 03:08:15**, four minutes after isolation, probably on a damaged flameproof light fitting.

A **Level-III emergency** was declared at about 03:10 and mutual aid was called. **Active cooling began at about 03:50**, the fire was **extinguished at about 06:00**, and **all-clear came at 07:30**. No one inside the plant was injured. Fire water held because **three electric pumps** carried the load. **Three of the four diesel fire pumps were under maintenance or unreliable.** [EVD-1284; EVD-4286; EVD-8871; EVD-9612; EVD-4052; EVD-4507; EVD-6916; EVD-2432]

Five things for Fire Services

1. **The diesel fire-water backup was largely gone, and no one had diagnosed why.** Two pumps were under maintenance and one tripped after two minutes from an unrecorded cause. The night was saved by grid power staying on. *Ask: is degraded diesel capacity an impairment in our system?* [FS-1]
2. **Drill records held dates, not lessons.** Readiness could not be shown or improved. *Ask: do our records show what each drill taught and whether it was fixed?* [FS-2]
3. **The fire station had about 53 minutes of warning before ignition.** That is a fire-service phase: ignition control, staging, water curtains, accounting. Most plans begin at "fire". *Ask: what do our plans say we do before ignition?* [FS-3]

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4. **The night cannot be timed reliably.** Clocks differed by 2 to 12 minutes, the only camera was lost at ignition, and there was no fire-service log. *Ask: can we time our next response?* [FS-4]

5. **The strongest causes sat at the interfaces.** The ERDMP had no isolation authority, the valves were local only, detection only displayed an alarm, and warnings went through intermediate points. Fire Services can move these through ERDMP review, drills and joint inspection. [IF-1 to IF-6]

Three decisions for corporate Fire Services

Decision	Why	Evidence
Treat diesel fire-water capacity below design as a declared impairment at every installation, with compensating measures	Hazira's backup was one usable pump of four; the faults were known but not diagnosed or actioned	EVD-2432
Require every installation to drill a **large unignited release at night** within 12 months, timed against a synchronised clock	The pre-ignition window was the chance to prevent the fire; no record shows it was ever drilled	EVD-2130; EVD-4052
Ask each ERDMP owner to state, for major-release scenarios, **who may isolate without asking**, and to test it with Fire Services	46 minutes from detection to isolation; no delegated authority	EVD-4052; EVD-3932

What is in this pack

#	Document	For
02	**Fire Services Learnings**: full narrative, 4 fire-led and 6 interface learnings	Chief Fire Officers
03	**Drill Scenarios**: four exercises with Hazira baselines	Drill planners
04	**Readiness Indicators**: 20 leading indicators (13 owned, 7 interface)	Head of Fire Services

Source and limits. An independent, draft desk analysis by Eyekyam Risk Resolutions from publicly available information, using ProbeMatrix (HFACS-based, human-reviewed). It is not ONGC's investigation, it cannot see fire-service records, and corrections are welcome. Standards are cited as pointers only.

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